CMPU1018_CA1

Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score
Question 1	5/5
Question 2	2/2
Question 3	4/4
Question 4	3/3
Question 5	3/3
Question 6	1/1
Question 7	4/4
Total	22/22 (100%)

Well done, you survived CA1!

Performance Summary

Exam Name:	CMPU1018_CA1
Session ID:	1601330659838694
Student's Name:	Charlie Hughes
Exam Start:	Mon Nov 09 2026 09:00:22
Exam Stop:	Mon Nov 09 2026 09:24:21
Time Spent:	00:23:57

Created using Numbas (https://www.numbas.org.uk) v9.0, developed by Newcastle University (https://www.newcastle.ac.uk).

Question 1

This question will test your use of the Division Algorithm and the Extended Euclidean Algorithm.

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a)

Use the division algorithm to find the quotient q and the remainder r when dividing 49 by 13.

q = 3 ✔ Expected answer: 3 , r = 10 ✔ Expected answer: 10

▶ **Feedback**

Score: 2/2 ✔

b)

The gcd of 32 and 23 is 1 ✔ Expected answer: 1 .

▶ **Feedback**

Score: 1/1 ✔

c)

$gcd(32, 23) =$ -5 ✔ Expected answer: −5 $x32+$

7 ✔ Expected answer: 7 $x23$

▼ Feedback

Score: 2/2 ✔

1

✔ Your answer is correct. You were awarded **1** mark.

2

✔ Your answer is correct. You were awarded **1** mark.

You scored **2** marks for this part.

Question 2

Euclidean Algorithm

Apply the Euclidean algorithm to find the greatest common denominator between 2010 and 270. Show the steps.

Part a) Algorithm

Please enter all the steps of the Euclidean algorithm below (please enter the number of steps you need in the "Rows:" box.)

Rows: 3

$$\begin{pmatrix} 2010 = 270 \times 7 + 120 \\ 270 = 120 \times 2 + 30 \\ 120 = 30 \times 4 + 0 \end{pmatrix}$$ ✔

Expected answer: $$\begin{pmatrix} 2010 = 7 \times 270 + 120 \\ 270 = 2 \times 120 + 30 \\ 120 = 4 \times 30 + 0 \end{pmatrix}$$

▼ Feedback

Score: 1/1 ✔

✔ You were awarded **1** mark.

You scored **1** mark for this part.

Part b) gcd

gcd(2010,270)=

30 ✔

Expected answer: 30

▼ Feedback

Score: 1/1 ✔

✔ Your answer is correct. You were awarded **1** mark.

You scored **1** mark for this part.

Advice

In order to apply the Euclidean algorithm to find the gcd, you need to apply integer division repeatedly, until the remainder is 0.

Once you know the gcd between two numbers p and q you can find their lcm using the following formula:

$$\operatorname{lcm}(p,q) = \frac{p \times q}{\gcd(p,q)}$$

Question 3

Multiplication in modular arithmetic looks different to regular multiplication. Here is an example for multiplication in modulo 13.

×	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9	10	11	12
2	0	2	4	6	8	10	12	1	3	5	7	9	11
3	0	3	6	9	12	2	5	8	11	1	4	7	10
4	0	4	8	12	3	7	11	2	6	10	1	5	9
5	0	5	10	2	7	12	4	9	1	6	11	3	8
6	0	6	12	5	11	4	10	3	9	2	8	1	7
7	0	7	1	8	2	9	3	10	4	11	5	12	6
8	0	8	3	11	6	1	9	4	12	7	2	10	5
9	0	9	5	1	10	6	2	11	7	3	12	8	4
10	0	10	7	4	1	11	8	5	2	12	9	6	3
11	0	11	9	7	5	3	1	12	10	8	6	4	2
12	0	12	11	10	9	8	7	6	5	4	3	2	1

Use this multiplication table to answer following questions.

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a)

The number x such that $11x \equiv 1 \pmod{13}$ is called the **inverse** of 11 in $\mathbb{Z}_{13}$. Find the inverse of 11 in $\mathbb{Z}_{13}$.

6 ✔ Expected answer: 6

▼ Feedback

Score: 1/1 ✔

✔ $11 \times 6 = 1 \pmod{13}$. You were awarded **1** mark.

You scored **1** mark for this part.

b)

Solve $11x \equiv 8 \pmod{13}$ for x.

9 ✔ Expected answer: 9

▼ Feedback

Score: 1/1 ✔

✔ $11 \times 9 = 8 \pmod{13}$. You were awarded **1** mark.

You scored **1** mark for this part.

c)

Solve $11x \equiv 7 \pmod{13}$ for x.

3 ✔ Expected answer: 3

▼ Feedback

Score: 1/1 ✔

✔ $11 \times 3 = 7 \pmod{13}$. You were awarded **1** mark.

You scored **1** mark for this part.

d)

Solve $11x + 2 \equiv 10 \pmod{13}$ for x.

9 ✔ Expected answer: 9

▼ Feedback

Score: 1/1 ✔

✔ $11 \times 9 + 2 = 8 \pmod{13}$. You were awarded **1** mark.

You scored **1** mark for this part.

Advice

This question introduces the concept of multiplication in modular arithmetic. All the answers can be found by looking up the appropriate row and column in the given multiplication table.

Aside: It is easy to find the inverse of a when working in $\mathbb{Z}_p$ for some prime number p. By Fermat's Little Theorem $a^p \equiv a \pmod{p}$, so if a is not a multiple of p then $a^{-1} \equiv a^{p-2} \pmod{p}$.

Question 4

Given integers $a, b \in \mathbb{Z}$ we can write a in the form

$$a = qb + r$$

for some $q, r \in \mathbb{Z}$ where $0 \leq r < b$. The numbers q and r are called the **quotient** and **remainder** when a is divided by b:

$$\frac{a}{b} = q + \frac{r}{b}.$$

In **modular arithmetic** (in this case, modulo b) our interest is focused on the remainder. The operation $a \bmod b = r$.

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a)

Which numbers have a remainder of 4 when divided by 7? These numbers would be considered equal in modulo 7.

☐ -9 ☑ -10 ☑ -3 ☑ 11 ☐ 12 ☐ -4 ☑ 25

✔

Expected answer:

☐ -9 ☑ -10 ☑ -3 ☑ 11 ☐ 12 ☐ -4 ☑ 25

▼ Feedback

Score: 1/1 ✔

- ✔ $-10 = (-2) \times 7 + 4$ and so has remainder 4. You were awarded **0.25** marks.
- ✔ $-3 = (-1) \times 7 + 4$ and so has remainder 4. You were awarded **0.25** marks.
- ✔ $11 = 1 \times 7 + 4$ and so has remainder 4. You were awarded **0.25** marks.
- ✔ $25 = 3 \times 7 + 4$ and so has remainder 4. You were awarded **0.25** marks.

You scored **1** mark for this part.

b)

Which numbers are equal to 6 in modulo 9?

☐ 32 ☐ 16 ☑ 6 ☐ -4 ☑ 15 ☑ -12 ☑ -3

✔

Expected answer:

☐ 32 ☐ 16 ☑ 6 ☐ -4 ☑ 15 ☑ -12 ☑ -3

▼ Feedback

Score: 1/1 ✔

- ✔ $6 = (0) \times 9 + 6$ and so has remainder 6. You were awarded **0.25** marks.
- ✔ $15 = 1 \times 9 + 6$ and so has remainder 6. You were awarded **0.25** marks.
- ✔ $-12 = (-2) \times 9 + 6$ and so has remainder 6. You were awarded **0.25** marks.
- ✔ $-3 = (-1) \times 9 + 6$ and so has remainder 6. You were awarded **0.25** marks.

You scored **1** mark for this part.

c)

A number a is a **multiple** of b precisely when $a = qb$ for some number q. In terms of modular arithmetic we would say $a \bmod b = 0$ because a has no remainder when divided by b.

Enter a non-zero number which is congruent to $0 \pmod 7$ and also congruent to $0 \pmod 9$.

63 ✔

Expected answer: 63

▼ Feedback

Score: 1/1 ✔

$63 = 0 \pmod 7$.

$63 = 0 \pmod 9$.

✔ You were awarded **1** mark.

You scored **1** mark for this part.

Advice

The first two questions introduce the idea that different numbers in $\mathbb{Z}$ can be the same in modulo b. For example in modulo 7 the numbers

$$7, 14, 21, \cdots$$

are equivalent to zero. While the numbers

$$10, 19, 28, \cdots$$

are equivalent to $1 \pmod 9$.

The next question introduce a simple divisibility test: $a = qb = qb + 0$ exactly when $a \equiv 0 \pmod b$. This is just another way to say that b divides a, or that a is a multiple of b. In particular

$63 = 7 \times 9$ is divisible by 7 and 9.

Question 5

In each of these questions, write the given number as a product of powers of primes in ascending order.

Give your answer in list form, where

the 1st element represents the power of 2

the 2nd element represents the power of 3

the 3rd element represents the power of 5

the 4th element represents the power of 7

For example

$210 = 2 \times 3 \times 5 \times 7$ we would give our answer as $[1, 1, 1, 1]$

$63 = 3^2 \times 7$ we would give our answer as $[0, 2, 0, 1]$

$750 = 2 \times 3 \times 5^3$ we would give our answer as $[1, 1, 3, 0]$

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a)

$392 =$

`[3,0,0,2]` $[3, 0, 0, 2]$ ✔

Expected answer: `[ 3, 0, 0, 2 ]` $[3, 0, 0, 2]$

▼ Feedback

Score: 1/1 ✔

✔ Your answer is numerically correct. You were awarded **1** mark.

You scored **1** mark for this part.

b)

$42 =$

`[1,1,0,1]` $[1, 1, 0, 1]$ ✔

Expected answer: `[ 1, 1, 0, 1 ]` $[1, 1, 0, 1]$

▼ Feedback

Score: 1/1 ✔

✔ Your answer is numerically correct. You were awarded **1** mark.

You scored **1** mark for this part.

c)

$5880 =$

`[3,1,1,2]` $[3,1,1,2]$ ✔

Expected answer: `[ 3, 1, 1, 2 ]` $[3,1,1,2]$

▼ Feedback **Score: 1/1** ✔

✔ Your answer is numerically correct. You were awarded **1** mark.

You scored **1** mark for this part.

Advice

To find the prime factorisation of a number, keep breaking that number down into pairs of factors until all number written down are primes.

e.g. To find the prime factorisation of 2450 we could note that

$2450 = 245 \times 10$

$2450 = 245 \times \underline{5} \times \underline{2}$

$2450 = \underline{5} \times 49 \times \underline{5} \times \underline{2}$

$2450 = \underline{5} \times \underline{7} \times \underline{7} \times \underline{5} \times \underline{2}$

where we have underlined the primes as we go along.

Rewriting this in ascending order of primes, we obtain $2450 = 2 \times 5 \times 5 \times 7 \times 7$

which we can also write as $2450 = 2 \times 5^2 \times 7^2$

(a) Applying a similar method to the given questions, we can obtain:

We can write the given numbers as products of prime factors as follows:

$392 = 2^3 \times 7^2$

$42 = 2 \times 3 \times 7$

$5880 = 2^3 \times 3 \times 5 \times 7^2$

Question 6

Find α and β.

Using the rules of logs:

$$\log_a(x^b) = b\log(x)$$
$$\log_a(x) + \log_a(y) = \log_a(xy),$$
$$\log_a(x) - \log_a(y) = \log_a\left(\frac{x}{y}\right).$$

Express the following in terms of $\log_a(x)$ and $\log_a(y)$

$$\log_a(x^{-5}y^4) = \alpha\log_a(x) + \beta\log_a(y)$$

$\alpha =$ -5 ✔ Expected answer: −5, $\beta =$ 4 ✔ Expected answer: 4

Gap 0

Score: 1/1 ✔

✔ Your answer is correct. You were awarded **0.5** marks.

Gap 1

✔ Your answer is correct. You were awarded **0.5** marks.

You scored **1** mark for this part.

Advice

The rules for combining logs are

$$\log_a(bc) = \log_a(b) + \log_a(c)$$

$$\log_a\left(\frac{b}{c}\right) = \log_a(b) - \log_a(c)$$

$$\log_a(b^r) = r\log_a(b)$$

a)
Using these rules gives:

$$\begin{aligned}\log_a(x^{-5}y^4) &= \log_a(x^{-5}) + \log_a(y^4)\\ &= -5\log_a(x) + 4\log_a(y)\end{aligned}$$

b)

$$\frac{21}{4}\log_a(x) + \log_a(6x-5) - \log_a(x^{\frac{1}{4}}) = \log_a(x^{\frac{21}{4}}) + \log_a(6x-5) - \log_a(x^{\frac{1}{4}})$$

$$\begin{aligned}&= \log_a\left(x^5(6x-5)\right)\\ &= \log_a\left(x^5(6x-5)\right)\end{aligned}$$

Question 7

a)

Express the following as a single power of 13:

$$13^8 \times 13^{12}$$

Enter the power 20 20 ✔ Expected answer: 20 20

▶ **Feedback**

Score: 1/1 ✔

b)

Express the following as a single power of 3:

$$(3^2)^7$$

Enter the power 14 14 ✔ Expected answer: 14 14

▶ **Feedback**

Score: 1/1 ✔

c)

Express the following as a power of a single number. The power should be a prime.

$$20^3 \times 23^3$$

Enter the number 460 460 ✔ Expected answer: 460 460 and the power 3 3 ✔ Expected answer: 3 3

▼ Feedback

Score: 2/2 ✔

Gap 0

✔ Your answer is numerically correct. You were awarded **1** mark.

Gap 1

✔ Your answer is numerically correct. You were awarded **1** mark.

You scored **2** marks for this part.

Advice

(a) Add the powers: $8 + 12 = 20$ (first index law).

(b) Multiply the powers $2 \times 7 = 14$ (second index law).

(c) Multiply 20 and 23 to get 460, with the power unchanged at 3. [This is the only way for the power to be a prime.]

Created using Numbas (https://www.numbas.org.uk) v9.0, developed by Newcastle University (https://www.newcastle.ac.uk).